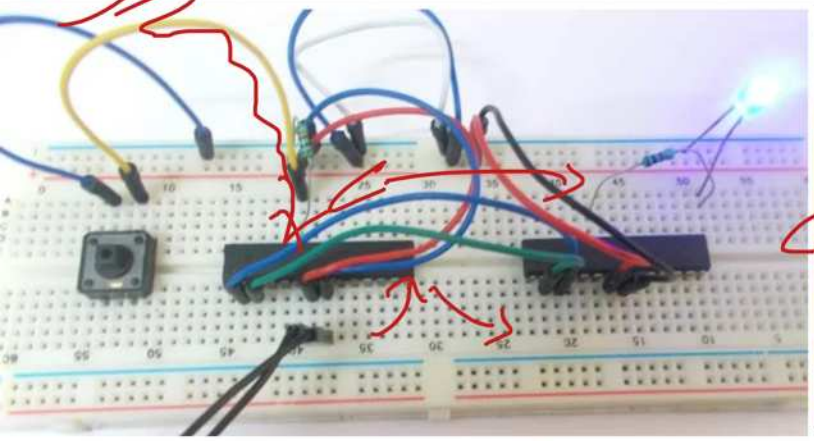
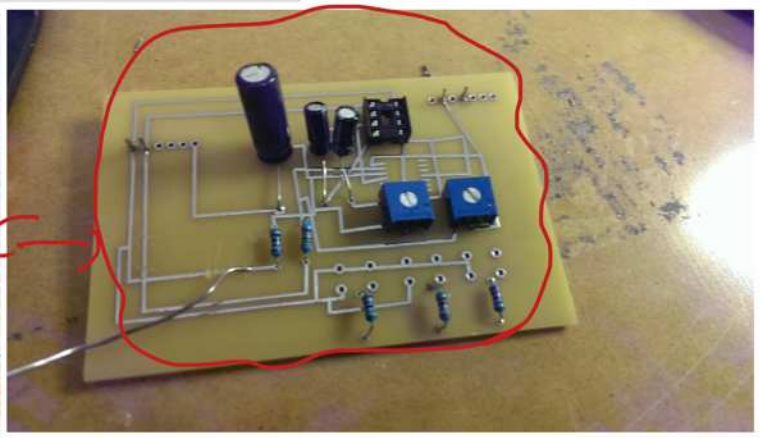

Device1
Data
CLK
Data
CLK
Device2








Logic
Power

Voyager's 15 Billion Mile Software Update

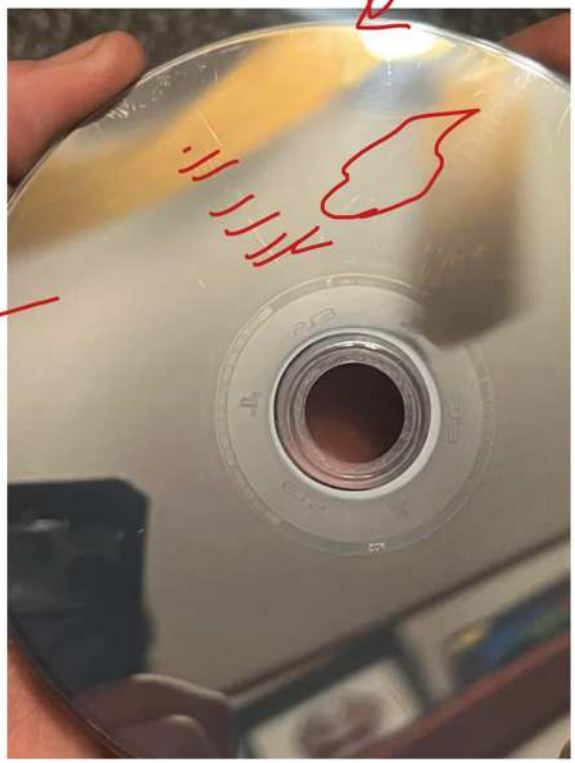


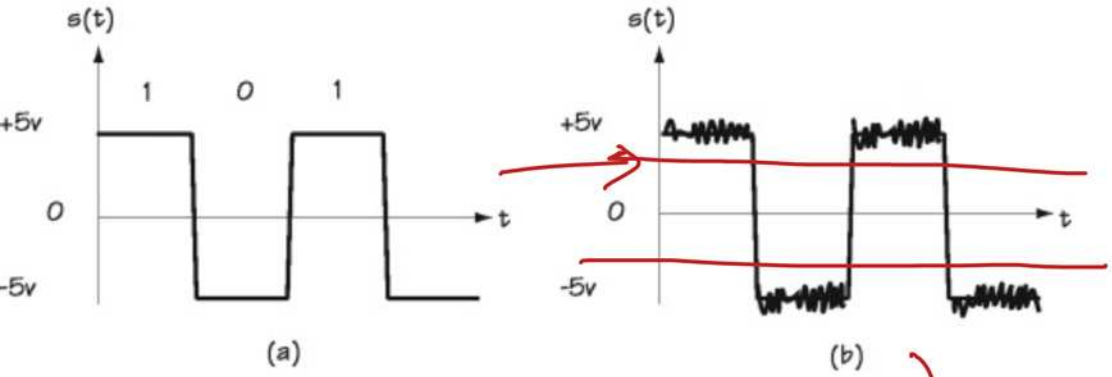
10

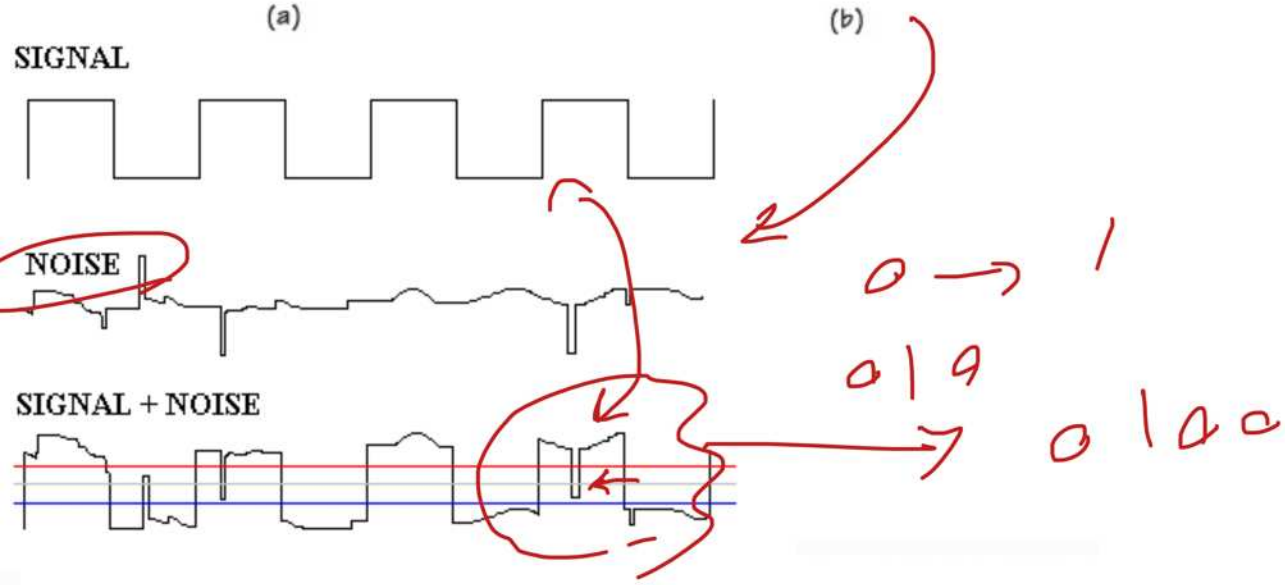


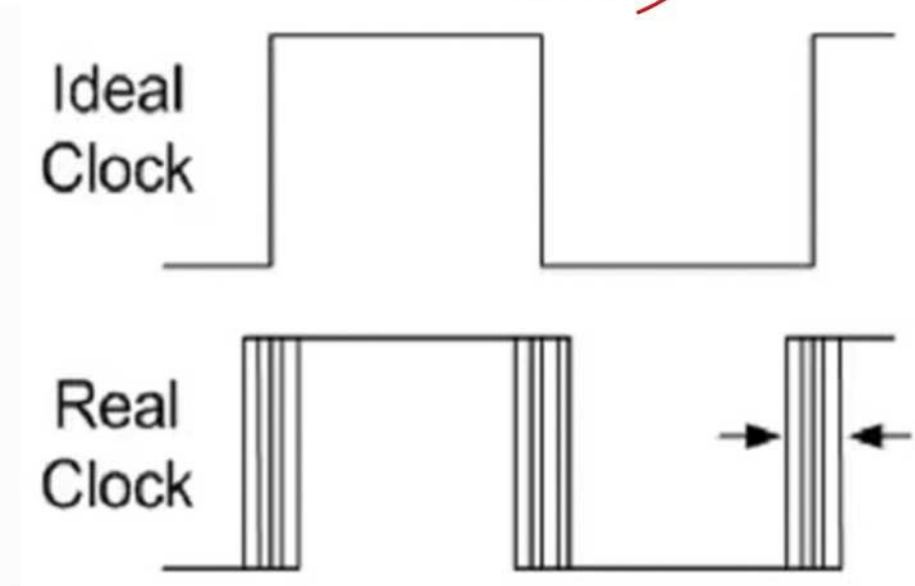


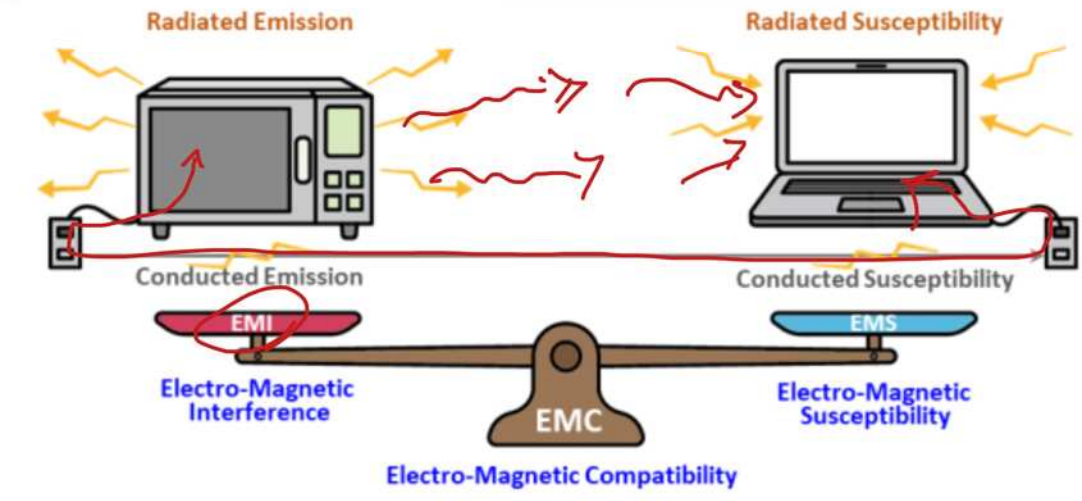





s(t)
1
0
1
+5v
0
-5v
t
(a)
s(t)
+5v
0
-5v
t
(b)

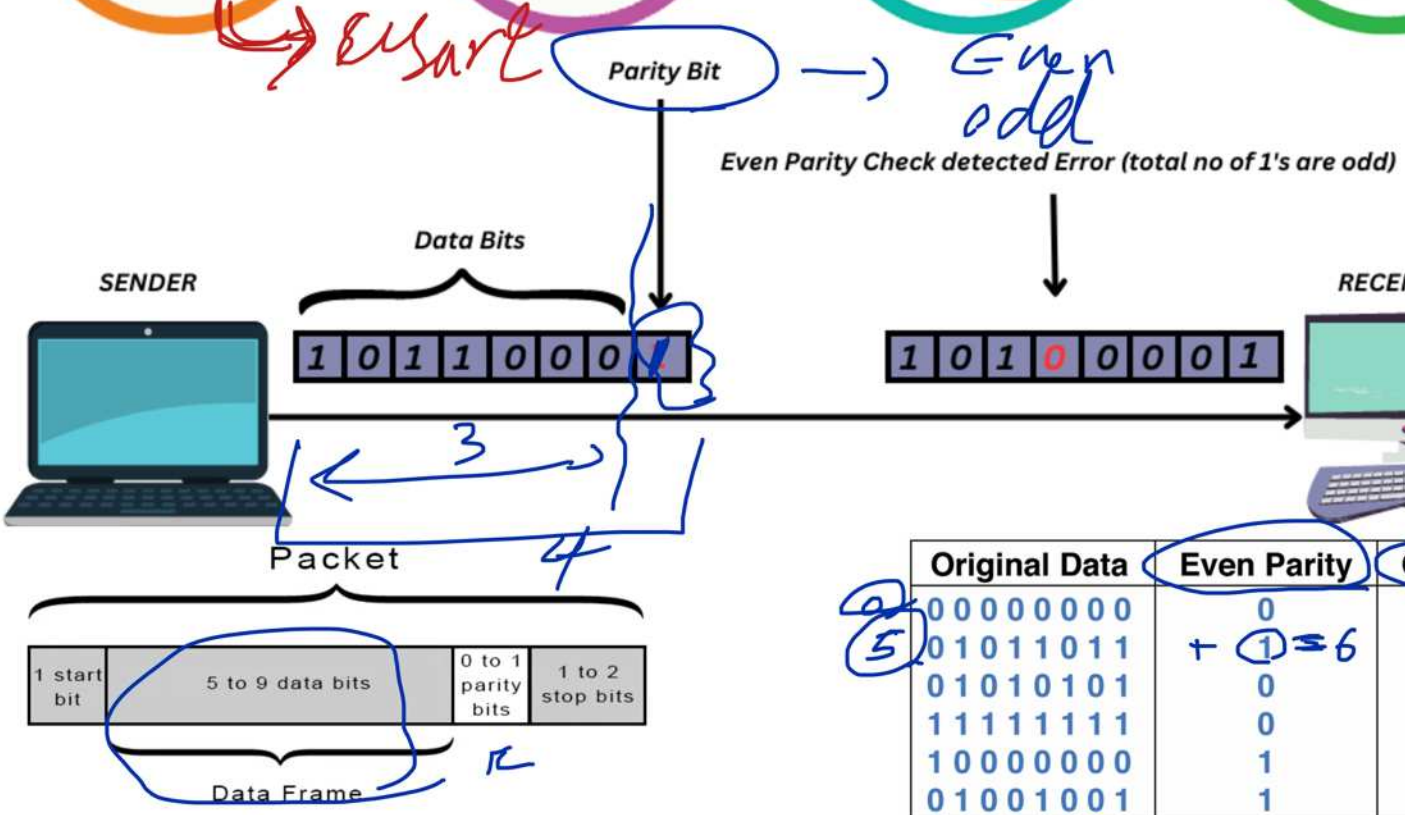

SIGNAL
NOISE
SIGNAL + NOISE


Ideal Clock
Real Clock
Jitter

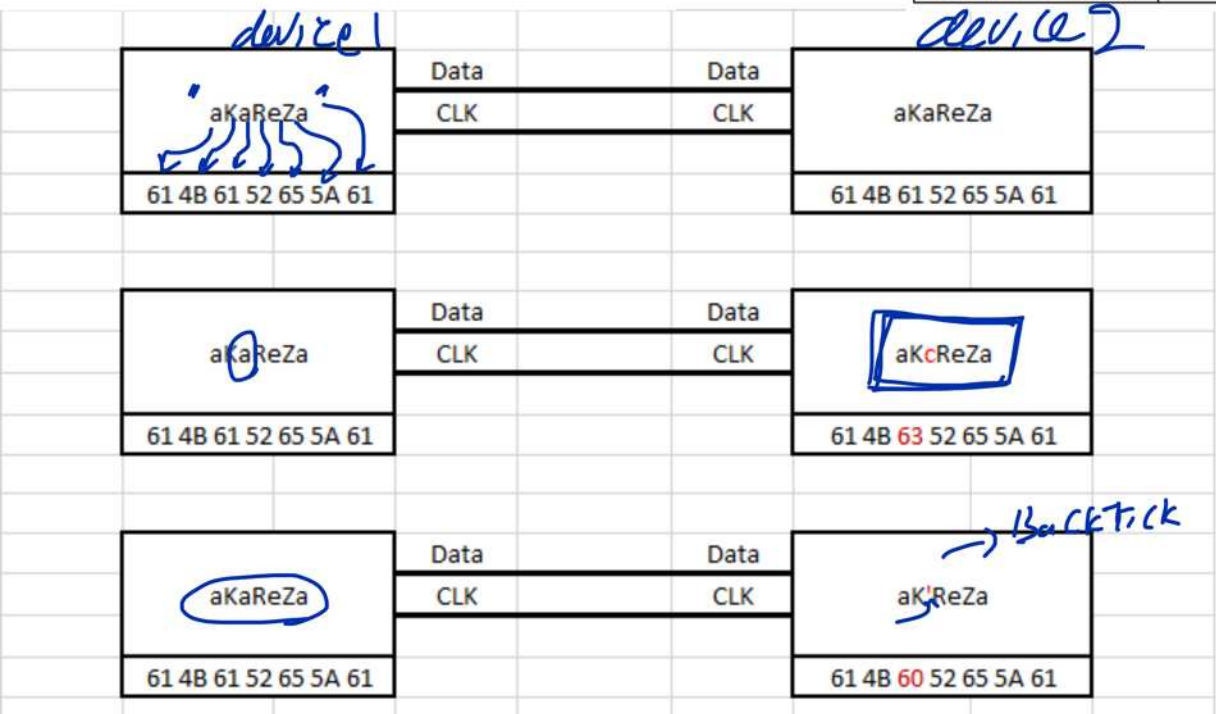

Radiated Emission
Radiated Susceptibility
Conducted Emission
Conducted Susceptibility
EMI
EMS
Electro-Magnetic Interference
EMC
Electro-Magnetic Susceptibility
Electro-Magnetic Compatibility

Error Detection and Correction Techniques

1. Parity Code
2. Checksum
3. Cyclic Redundancy Check(CRC)
4. Hamming Code



Original Data	Even Parity	Odd Parity
0 0 0 0 0 0 0 0	0	1
0 1 0 1 1 0 1 1	1	0
0 1 0 1 0 1 0 1	0	1
1 1 1 1 1 1 1 1	0	1
1 0 0 0 0 0 0 0	1	0
0 1 0 0 1 0 0 1	1	0



Char	Hex	Dec	Bin	Num 1	Parrity
a	0x61	97	0b 0110 0001	3+1	1
K	0x4B	75	0b 0100 1011	4+0	0
a	0x61	97	0b 0110 0001	3+1	1
R	0x52	82	0b 0101 0010	3+1	1
e	0x65	101	0b 0110 0101	4+0	0
Z	0x5A	90	0b 0101 1010	4+0	0
a	0x61	97	0b 0110 0001	3+1	1

Parity Even

Parity Even

aKaReZa	Data / CLK	Data / CLK	aKcReZa
61 4B 61 52 65 5A 61			61 4B 63 52 65 5A 61

Char	Hex	Dec	Bin	Num 1	Parrity
a	0x61	97	0b 0110 0001	3+1	1
c	0x63	99	0b 0110 0011	4+0	0

Parity

0x61 | 1

1

0x63 | 1 X

Parity

aKaReZa	Data / CLK	Data / CLK	aK'ReZa
61 4B 61 52 65 5A 61			61 4B 60 52 65 5A 61

Char	Hex	Dec	Bin	Num 1	Parrity
a	0x61	97	0b 0110 0001	3	1
backtick	0x60	96	0b 0110 0000	2+0	0

0x6c | 1

0 X

aKaReZa	Data / CLK	Data / CLK	aKfReZa
61 4B 61 52 65 5A 61			61 4B 66 52 65 5A 61

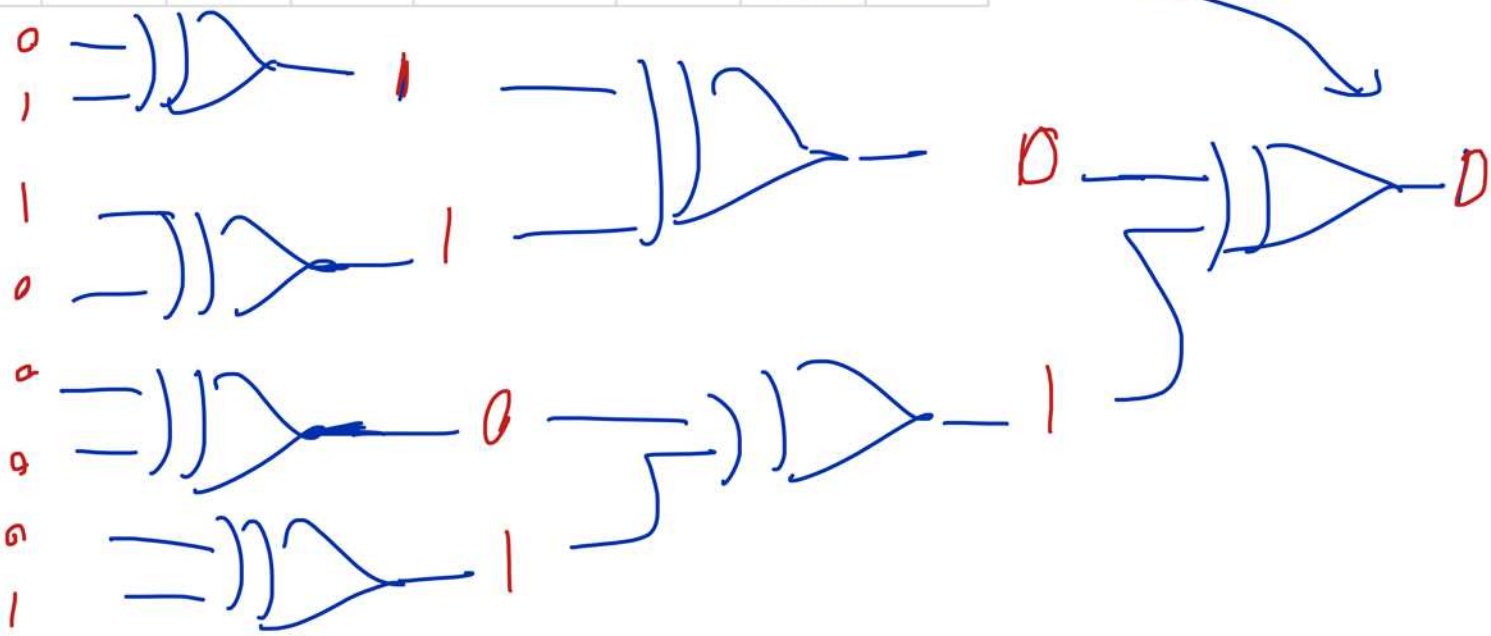
Char	Hex	Dec	Bin	Num 1	Parity
a	0x61	97	0b 0110 0001	3	1
f	0x66	102	0b 0110 0110	4+0	0

0x66 | 1

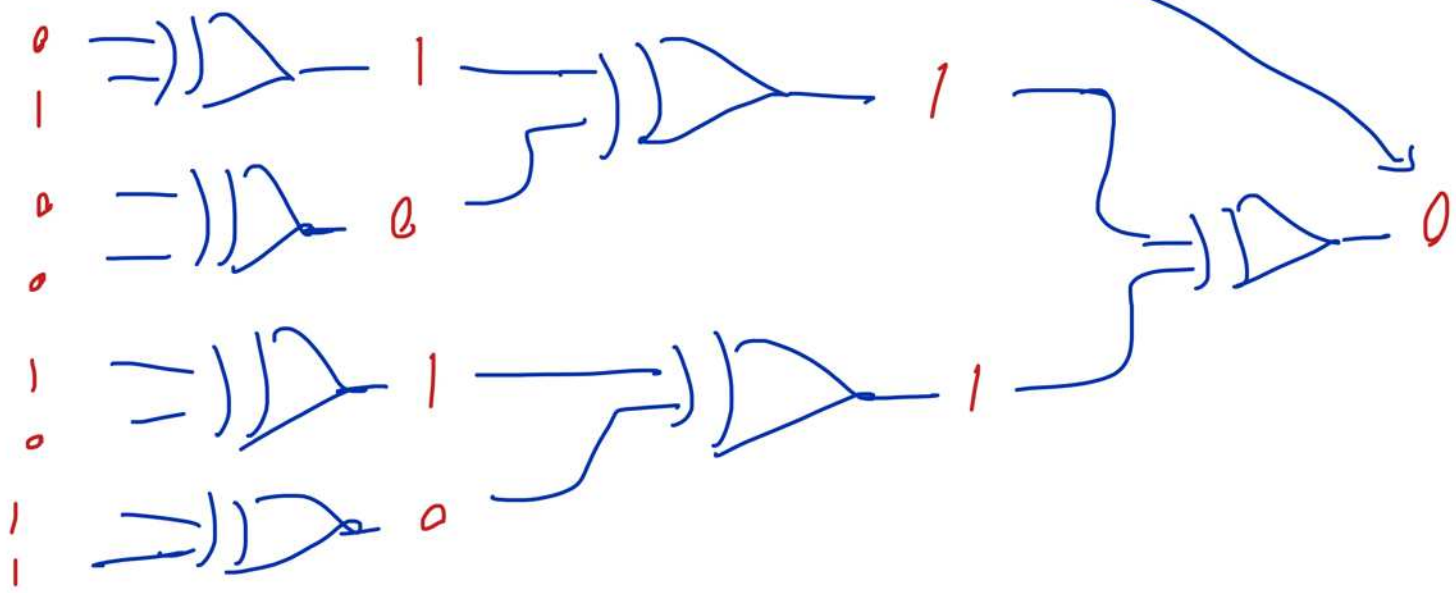
q q

Char	Hex	Dec	Bin	Num 1	Parrity
a	0x61	97	0b 0110 0001	3	1

Even



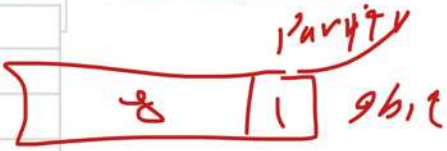
Char	Hex	Dec	Bin	Num 1	Parrity
K	0x4B	75	0b 0100 1011	4	0



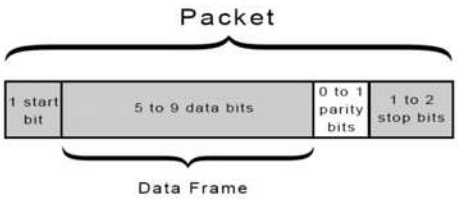
aKaReZa	Data / CLK	Data / CLK	aKbReZa
61 4B 61 52 65 5A 61			61 4B 62 52 65 5A 61

Char	Hex	Dec	Bin	Num 1	Parrity
a	0x61	97	0b 0110 0001	3	1
b	~~0x5B~~ 0x62	~~88~~ 98	0b 0110 0010	3+1	1











Sum = 97 + 75 + 97 + 82

Sum = 411

411 + 99 + 97 = 639

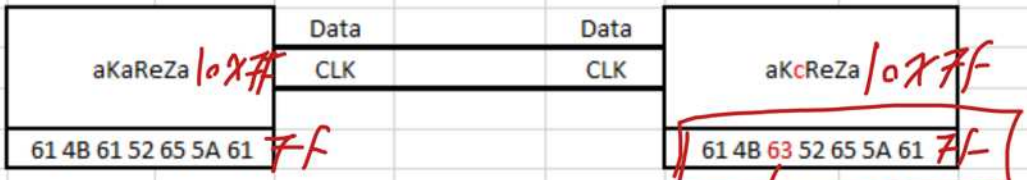
Char	Hex	Dec	Bin	Num 1	Parity
a	0x61	97	0b 0110 0001	3	1
K	0x4B	75	0b 0100 1011	4	0
a	0x61	97	0b 0110 0001	3	1
R	0x52	82	0b 0101 0010	3	1
e	0x65	101	0b 0110 0101	4	0
Z	0x5A	90	0b 0101 1010	4	0
a	0x61	97	0b 0110 0001	3	1

639 % 256 = 127

Checksum = 0x7f

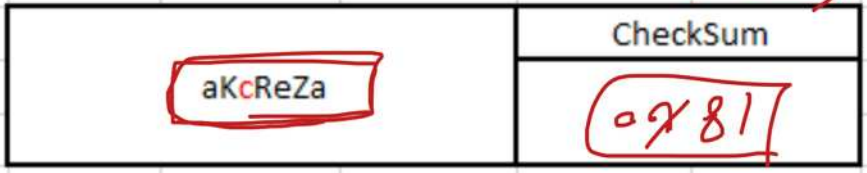
aKaReZa	a	K	a	R	e	Z	a
Hex	0x61	0x4B	0x61	0x52	0x65	0x5A	0x61
Dec	97	75	97	82	101	90	97
Bin	01100001	01001011	01100001	1010010	01100101	01011010	01100001





99

691 % 256



≠ 0x7f

9 ⟷ x



aKaReZa	CheckSum
61 4B 61 52 65 5A 61	0x7F

aZeRaKa	CheckSum
61 5A 65 52 61 4B 61	0x7F

aKaReZa	CheckSum
61 4B 61 52 65 5A 61 00 00 00	0x7F

aKaReXc	CheckSum
61 4B 61 52 65 [illegible] 58 63	0x7F

a+b+C
=
C+b+a
=
a+C+b

a+a = a

X Hamming Distance

= 2

Char	Hex	Dec	Bin	Num 1	Parity
Z	0x5A	90	0b 0101 1010	4	0
X	0x58	88	0b 0101 1000	3	1
a	0x61	97	0b 0110 0001	3	1
c	0x63	99	0b 0110 0011	4	0

Column

2

Row

100

Row

8 bit

Row

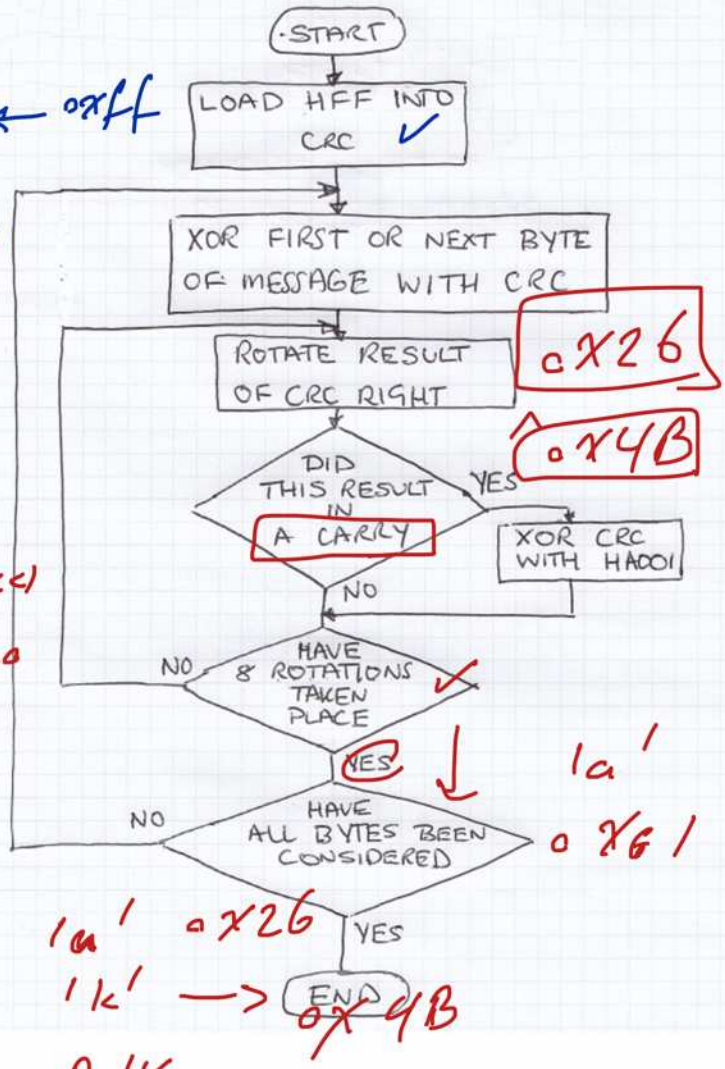
Char	b7	b6	b5	b4	b3	b2	b1	b0	No 1	Parity
a	0	1	1	0	0	0	0	1	3	1
K	0	1	0	0	1	0	1	1	4	0
a	0	1	1	0	0	0	0	1	3	1
R	0	1	0	1	0	0	1	0	3	1
e	0	1	1	0	0	1	0	1	4	0
Z	0	1	0	1	1	0	1	0	4	0
a	0	1	1	0	0	0	0	1	3	1
No 1	0	7	4	2	2	1	3	5		
Parity	0	1	0	0	0	1	1	1		

Char	b7	b6	b5	b4	b3	b2	b1	b0	No 1	Parity
a	0	1	1	0	0	0	0	1	3	1
K	0	1	0	0	1	0	1	1	4	0
a	0	1	1	0	0	0	0	1	3	1
R	0	1	0	1	0	0	1	0	3	1
e	0	1	1	0	0	1	0	1	4	0
X	0	1	0	1	1	0	0	0	3	1
c	0	1	1	0	0	0	1	1	4	0
No 1	0	7	4	2	2	1	3	5		
Parity	0	1	0	0	0	1	1	1		



CRC-8

CRC FLOW DIAGRAM



0xff

0x26

0x4B

'a' 0x61

'a' 0x26

'k' -> 0x4B

0	CRC	1	1	1	1	1	1	1	1	0xFF

1	CRC	1	1	1	1	1	1	1	1	0xFF
+	a	0	1	1	0	0	0	0	1	0x61
!	CRC	1	0	0	1	1	1	1	0	0x9E

Carry

CRC = CRC ^ 0x61

CRC = CRC<<1;

	CRC	0	0	1	1	1	1	0	0	0x3C
	Poly	0	0	1	1	0	0	0	1	0x31
	CRC	0	0	0	0	1	1	0	1	0x0D

CRC = CRC ^ 0x31;

0x0D <<1

0	CRC	0	0	0	1	1	0	1	0	0x1A
0	CRC	0	0	1	1	0	1	0	0	0x34
0	CRC	0	1	1	0	1	0	0	0	0x68

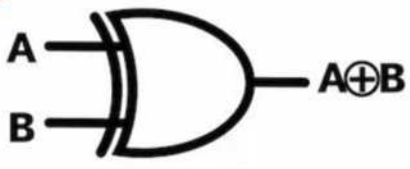
msb

0	CRC	1	1	0	1	0	0	0	0	0xD0

CRC = 0xA0 ^ 0x31

1	CRC	1	0	1	0	0	0	0	0	0xA0
	Poly	0	0	1	1	0	0	0	1	0x31
	CRC	1	0	0	1	0	0	0	1	0x91

Poly



*eXclusive OR

2 input XOR gate

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

1	CRC	0	0	1	0	0	0	1	0	0x22
	Poly	0	0	1	1	0	0	0	1	0x31
	CRC	0	0	0	1	0	0	1	1	0x13

CRC = 0x22 ^ 0x31

CRC-8/NRSC-5	
Initial Value	0xFF
Polynomial	0x31
Input reflected	FALSE
Result reflected	FALSE
Final XOR	0x00

AMT20

0x26

	CRC	0	0	1	0	0	1	1	0	0x26
	xor	0	0	0	0	0	0	0	0	0x00
	CRC	0	0	1	0	0	1	1	0	0x26

CRC = 0x26 ^ 0x00

=> CRC = 0x26

